

|           |          |                                                                                                                                                                                                                                                                                                                            |
|-----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           |          | proportionally with $I$ .                                                                                                                                                                                                                                                                                                  |
| <b>21</b> | <b>D</b> | <p>From <math>I = Anvq = \left( \frac{\pi d^2}{4} \right) nvq</math> where <math>d</math> = diameter</p> <p><math>\rightarrow v = \frac{4I}{\pi d^2 nq} \rightarrow v \propto \frac{1}{d^2}</math></p> <p>As the wire tapers from P to Q, its diameter decreases uniformly, so <math>v</math> increases non-uniformly.</p> |
| <b>22</b> | <b>A</b> | If the broken third bulb is the only fault, the voltmeter would show the reading of                                                                                                                                                                                                                                        |